

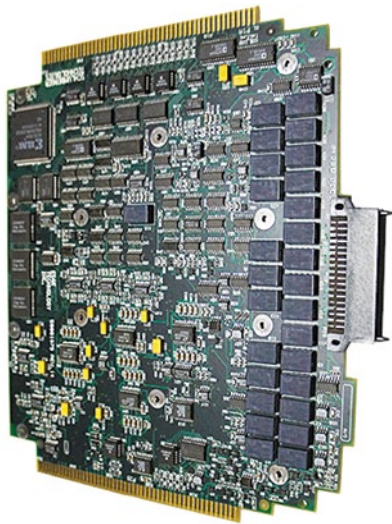


Fig. 2: DAQ board from Analog Arts
(Image courtesy: www.analogarts.com)

Make way for better processing...

How best to convert the available analogue data to a digital signal? We now have Sigma-Delta analogue-to-digital converters (ADCs) that work with 24-bit data, which demand over-sampling. Filters onboard the ADC or its card strengthen the signal to make it suitable for long-distance transmission.

This method is slowly making its mark in the industry, owing to high measurement accuracy. It is also useful when transferring high bit-count low-frequency digital signals into lower bit-count higher-frequency digital signals, during the reverse process. "New signal-processing components offer high resolution at extremely-low power consumption," Ushani says.

... Analyse better and store better

Better the processing, better the analysis. Naturally, companies are vying with one another to introduce advanced processing techniques. "Today, most DAQ companies have front-end analysis as part of the large data tools being sold. National Instruments has Diadem, a third-party software available to pull up data and look at graphs, or pre-measure to see the effects of one on top of the other. These

Will USB continue to rule data transfer? Let us wait and watch

Shayan Ushani, business developer, Analog Arts, says, "USB-type products are gaining popularity among users. At Analog Arts, we believe that USB DAQ will be the most popular form of DAQ in the near future. The ease of use makes it ideal for data transfer for a majority of applications. Data transfer rate of the USB port was a major issue for some high-end applications. However, USB 3.0 has taken away this weakness."

Contrary to this, Vinod Mathews, founder and CEO, Captronics Systems Pvt Ltd, says "Maximum systems are running on USB and a personal computer is becoming a closed unit. USB 3.0 has no more future and will be taken over by faster interfaces, starting with Wi-Fi itself. Once Wi-Fi is faster, USB becomes irrelevant."

Re-usable electronics on wings

In the words of Vinod Mathews, founder and CEO, Captronics Systems Pvt Ltd, "Now, Polar Satellite Launch Vehicle launches are one-tenth the cost of National Aeronautics and Space Administration (NASA) and European Space Agency (ESA) launches. If they develop electronics and vehicles that come back, that is bound to increase the level of competition in India. Electronics that comes back can be re-used to cut down the cost. This will be the programme of the future."

This is in sync with one of their earlier projects where they needed to acquire data from the capsule in a parachute, while it was at the horizon. They could get the data once the parachute landed. This concept can be extended to the defence sector, Sound Navigation and Ranging (SONAR) technology and more. "Hence, for the future, Captronics is working on design issues and with companies who are working on ruggedising the systems," adds Mathews.

are in an easy-to-use format and do not demand prior knowledge of programming. You can just open up the sample in the software. It pulls in the data and you can throw it around," comments Mathews.

In addition to grabbing maximum data from the logged data, the newer range of algorithms allow for better data compression and storage. With these algorithms being built into DAQ systems, there is more storage space for data and, more importantly, more data can be stored in lesser space, in Mathews' view.

Why worry about space when you have the cloud

Although cloud processing has not caught on much when it comes to DAQs, there is a practice of storing data in the cloud, avoiding storage-and-memory problems. Mathews opines, "Data is digitised and we can have a buffer of a few GB onboard, which makes it easy, and I can then transfer whatever I want to send to a server. Basically, the cloud is used for the back-end. The person who is using the data need not be sitting on the factory floor. Today, he or she

can walk through the factory with a tablet and get all the information that is required, because they are all connected through Wi-Fi."

However, more connected the system is, more is the lurking danger. "Data privacy and security are becoming extremely important factors of the new generation of DAQ products," mentions Ushani.

"The reason for this new-found importance is data sharing through the Internet. In addition to this, remote control of these devices for some applications demands a high degree of security," he adds. Strictly following standards and protocols seems to be the best way to keep the data safe.

Interesting developments

With all these developments, allowing for equipment to be configurable creates the much-needed flexibility for user applications. Varun Manwani, director, Sahasra Electronics, whose company is the exclusive distributor of Labjack products, says, "Devices from Labjack are plug-and-play that do not need a whole lot of time to understand, install and use. These work on open source soft-